

EDUCATION

Doctor of Philosophy | Neuroscience | University of Victoria | Victoria, BC 2015 - 2020
Doctor of Medicine | Universidad del Valle, School of Medicine | Cali, Colombia 2007 - 2014

RELEVANT CLINICAL EXPERIENCES

Observership: Functional Neurosurgery - 4 weeks | Supervisor: Dr. Konstantin Slavin 2014
• University of Illinois Hospital & Health Sciences System, Chicago IL

Pre-Diploma Internship (*Internado Rotatorio*) 2013 - 2014
• Hospital Universitario del Valle ESE | Universidad del Valle, School of Medicine | Cali, Colombia
◦ Internal Medicine (*Adulto y Anciano III*) - 7 weeks
◦ Paediatrics (*Niño y Adolescente III*) - 7 weeks
◦ Obstetrics and Gynaecology (*Salud de la Mujer III*) - 7 weeks
◦ Specialized Research Internship in Biomedical Science (*Neuroscience Research*) - 16 weeks
◦ Surgery (*Trauma y Violencia III*) - 7 weeks
◦ Family Medicine (*Medicina Familiar*) - 4 weeks

Professional Elective: Neurosurgery (*Neurocirugía*) - 4 weeks 2012
• Hospital Universitario del Valle ESE | Universidad del Valle, School of Medicine | Cali, Colombia

Professional Elective: Systems Pathology (*Patología de Sistemas*) - 4 weeks 2011
• Universidad del Valle, School of Medicine | Cali, Colombia

SELECTED AWARDS

Research Trainee Award | Michael Smith Health Research BC 2021 - 2022
Nb Gulila Star Award | International Gap Junction Conference 2019
John & Myrtle Tilley Graduate Scholarship | University of Victoria, Faculty of Graduate Studies 2019
James & Laurette Agnew Memorial Scholarship | University of Victoria, Faculty of Graduate Studies 2015 - 2019
Donald Wagg Graduate Scholarship | University of Victoria, Faculty of Graduate Studies 2017 - 2018
Graduate Award | University of Victoria, Faculty of Graduate Studies 2015 - 2017
Fellowship Entrance Award | University of Victoria, Faculty of Graduate Studies 2015
School of Medicine Yearly Dean's List | Universidad del Valle, School of Medicine 2007, 2009, 2013, 2014

RESEARCH EXPERIENCE

Postdoctoral Research Fellow | University of Victoria, Division of Medical Sciences | Victoria, BC 2020 - 2022
• Led the design of a holistic and culturally safe translational research project to investigate an arrhythmia-causing genetic variant identified in the Gitksan First Nation.
• Managed complex genomic and transcriptomic datasets using advanced bioinformatics (R, Bioconductor) to discover mechanisms underlying cardiomyocyte and neuronal calcium homeostasis.
• Communicated vital research outcomes via peer-reviewed journals and community-based talks, enhancing the collective knowledge of congenital cardiac conditions.

Graduate Research Assistant | University of Victoria, Division of Medical Sciences | Victoria, BC 2015 - 2020
• Engineered conditional knockout mouse models and optimized primary neuronal cultures to investigate mechanisms of dendritic spine plasticity and cerebral cortex development.
• Developed advanced microscopy methods to visualize cellular dynamics.
• Championed open-science practices by establishing accessible bio-image analysis frameworks for the research community.

Undergraduate Medical Researcher | Universidad del Valle, Centro de Estudios Cerebrales | Cali, Colombia 2014

- Facilitated literature seminars and thematic discussions for researchers and partners at the Centro de Estudios Cerebrales.
- Refined experimental procedures for modeling diffuse traumatic brain injury and developed staining techniques for organotypic brain tissue preparations.

Undergraduate Medical Researcher | Universidad del Valle, School of Public Health | Cali, Colombia 2010 - 2011

- Coordinated a regional survey assessing cardiovascular risk factors in spinal cord injury patients, identifying key social determinants of health to inform targeted rehabilitation plans.
- Collaborated with Physical Medicine and Rehabilitation residents and attendings to prescribe comprehensive, evidence-based management plans for chronic patients.

MEDICAL EDUCATION & TEACHING EXPERIENCE

Teaching Assistant | Island Medical Program, UBC | University of Victoria | Victoria BC 2017 - 2020

Foundations of Medical Practice I and II (MEDD412, MEDD421, MEDD422)

- Instructed first- and second-year medical trainees in neuroanatomy laboratory sessions, fostering a dynamic educational space centered on the functional integration of clinical signs with neural structures and physiological processes.

LEADERSHIP & COMMUNITY ENGAGEMENT

Builder/Hacker | Placement: Third Place | Island Health Code Hack | Victoria, BC 2023

- Collaborated within an interdisciplinary team to pitch and develop targeted technological solutions aimed at reducing "PJ paralysis" and preventing patient deconditioning across Island Health.

Co-creator & Organizer | "Who can become a scientist?" Symposium | Victoria, BC 2021 - 2022

- Established an educational symposium for high school students addressing equity, diversity, and inclusion in STEM, advocating for underrepresented populations in the sciences.

Local Chapter Coordinator | CIHR Brain Bee | University of Victoria | Victoria, BC 2017 - 2023

- Directed local operations and social media for the Victoria Brain Bee, mentoring high school students and securing funding to sponsor winners for national competitions.

Newsletter Co-editor | BC Regenerative Medicine Network | Victoria, BC 2017 - 2019

- Curated and edited organizational communications, recruited symposium speakers, and evaluated scientific abstracts to promote regional collaboration in regenerative medicine.

Student Representative | Neuroscience Graduate Program | University of Victoria | Victoria, BC 2017 - 2018

- Spearheaded collaborative programs dedicated to championing psychological wellness and mental health support for graduate trainees.
- Orchestrated the recruitment of distinguished researchers for the Neuroscience Graduate Program Seminar Series, facilitating meaningful student engagement with global experts.

WORK EXPERIENCE

Data Scientist | BenchSci | Toronto, ON (Remote) 2025 - Present

- Architecting AI-driven data intelligence pipelines and hardening extraction systems to deliver high-throughput API endpoints to support agentic preclinical biomedical acceleration.

Science Tech Lead and Team Lead | BenchSci | Toronto, ON (Remote) 2023 - 2025

- Led multidisciplinary teams and directed technical strategy to improve the observability and scientific quality of data pipelines.
- Engineered comprehensive analytics frameworks to evaluate disease relations across journal articles (MONDO, ICD-10, SNOMED), ensuring robust data quality metrics and standards.

- Partnered with cross-functional stakeholders (Product, Engineering) to align scientific requirements with shared technical execution and delivery.

Scientist II - Data Extraction | BenchSci | Toronto, ON (Remote) 2022 - 2023

- Developed and optimized biomedical data extraction workflows to enhance precision and data quality.

PROFESSIONAL DEVELOPMENT

Google Data Analytics Specialization, Google Cloud	2026
Certified Google Generative AI Leader, Google Cloud	2025
Certified Scrum Product Owner (1920762), Scrum Alliance	2024
Course in Genetic Epidemiology and Epigenetics, Universidad del Valle, School of Public Health	2022
Code in Place, Stanford University	2021

PROFESSIONAL AFFILIATIONS

MEMBER Colegio Colombiano de Neurociencias (COLNE)	2020 - Ongoing
MEMBER Canadian Cardiovascular Society (CSS)	2021 - 2022
MEMBER Canadian Association for Neuroscience (CAN)	2015 - 2022
MEMBER Society for Neuroscience (SFN)	2016 - 2020

SELECTED PEER-REVIEWED PUBLICATIONS (Additional entries available in [Google Scholar profile](#))

York, Nicole S; **Sanchez-Arias, Juan C**; McAdam, Alexa CH; Rivera, Joel E; Arbour, Laura T; Swayne, Leigh Anne. *Mechanisms underlying the role of ankyrin-B in cardiac and neurological health and disease*. Frontiers in Cardiovascular Medicine 9, 964675. **2022**. Frontiers Media SA.

Kallas, Dania; Bos, Johan Martijn; Peltenburg, Puck; Franciosi, Sonia; Mohajeri, Arezoo; Cheung, Matthew; Brett, Laura; York, Nicole; **Sanchez-Arias, Juan C**; Swayne, Leigh Anne. *PO-655-05 IS ANKYRIN-2 A POTENTIAL GENETIC MODIFIER IN PEDIATRIC CATECHOLAMINERGIC POLYMORPHIC VENTRICULAR TACHYCARDIA?*. Heart Rhythm 19,5,S264-S265. **2022**. Elsevier.

Sanchez-Arias, Juan C; Carrier, Micael; Frederiksen, Simona D; Shevtsova, Olga; McKee, Chloe; van der Slagt, Emma; Gonçalves de Andrade, Elisa; Nguyen, Hai Lam; Young, Penelope A; Tremblay, Marie-eve. *A systematic, open-science framework for quantification of cell-types in mouse brain sections using fluorescence microscopy*. Frontiers in Neuroanatomy. 15, 722443. **2021**. Frontiers Media SA.

Sanchez-Arias, Juan C; van der Slagt, Emma; Vecchiarelli, Haley A; Candlish, Rebecca C; York, Nicole; Young, Penelope A; Shevtsova, Olga; Juma, Afnan; Tremblay, Marie-Eve; Swayne, Leigh Anne. *Purinergic signaling in nervous system health and disease: Focus on pannexin 1*. Pharmacology & therapeutics 225, 107840. **2021**. Pergamon.

Sanchez-Arias, Juan C; Candlish, Rebecca C; van der Slagt, Emma; Swayne, Leigh Anne. *Pannexin 1 regulates dendritic protrusion dynamics in immature cortical neurons*. ENeuro,7,4,. **2020**. Society for Neuroscience.

Swayne, Leigh Anne; Johnstone, Scott R; Ng, Chen Seng; **Sanchez-Arias, Juan C**; Good, Miranda E; Penuela, Silvia; Lohman, Alexander W; Wolpe, Abigail G; Laubach, Victor E; Koval, Michael. *Consideration of Pannexin 1 channels in COVID-19 pathology and treatment*. American Journal of Physiology-Lung Cellular and Molecular Physiology 319,1,L121-L125. **2020**. American Physiological Society Bethesda, MD.

Sanchez-Arias, Juan C; Liu, Mei; Choi, Catherine SW; Ebert, Sarah N; Brown, Craig E; Swayne, Leigh Anne. *Pannexin 1 regulates network ensembles and dendritic spine development in cortical neurons*. Eneuro, 63. **2019**. Society for Neuroscience.

HOBBIES & INTERESTS

Sports: Strength conditioning, football, road cycling | **Technology:** Hackathons, Free Open Source Software Development, AI-driven Software Development

LANGUAGES

English: Fluent | **Spanish:** Native